Memory Interpreter

Parameters

- ADDR_WIDTH.
- MEM_SIZE = 512.
- ADDR_SIZE = 9.
- RANGES_NUMBER = 2.

Input Signals

Signal	Number of bis	Description
end_range	1	When high, it indicates that the sent address and data is the las in this range
address_in	ADDR_WIDTH	It's the logical address for the output data that should come from the RAM
change_direction	1	When high, you must change the direction, if you were incrementing the address decrement and vice versa
en	1	When high the saved address is incremented or decremented when low the saved address is constant

Output Signals

Signal	Number of bits	Description
out_address_out	ADDR_SIZE	The physical address for the output data that should come from the RAM
alu_address_out	ADDR_SIZE	The physical address for the alu data that should come from the RAM
output_valid	1	High when the address_in contains ready data